Accio

Fabrication Manual: Zenith Aero Bamboo Pedicab (The Voyager)

This technical guide provides the specifications and manual construction steps required to build Proposal 2: The Voyager — Luxe Laminated Concept. This vehicle utilizes heat-treated laminated bamboo to achieve a high-strength, lightweight chassis.

1. Fabrication of the Lamination Jigs ("Blue Boards")

The "Blue Boards" represent the CNC-cut templates used as molds for shaping the curved bamboo components. These should be cut from 18mm plywood or MDF.

Cutting List for Templates

Component ID	Description	Template Length	Function
BB-01	Main Chassis Rail (Left/Right)	2400 mm (94.5")	Primary structural curve
BB-02	Wheel Arch Support	1200 mm (47.2")	Rear axle mounting curve
BB-03	Passenger Bench Frame	1300 mm (51.2")	Seat structure support
BB-04	Steering Column Strut	900 mm (35.4")	Front fork connection

Instruction: Cut two of each template to allow for parallel lamination of symmetrical parts.

2. Bamboo Lamination Schedule

The frame is constructed by gluing and clamping thin bamboo strips (5mm thickness) into the jigs.

Section	Target Thickness	Layers of 5mm Strips	Notes
Main Chassis Rails	60 mm	12 Layers	Highest load section
Wheel Arches	50 mm	10 Layers	Structural reinforcement
Seat Supports	40 mm	8 Layers	Passenger weight load
Canopy Struts	20 mm	4 Layers	Aesthetic/Minor load

3. Joint Engineering: The "Overbuilt" Tongue & Tenon

To ensure organic flowing curves, joints are "overbuilt" and then hand-ground to their final ergonomic shape.

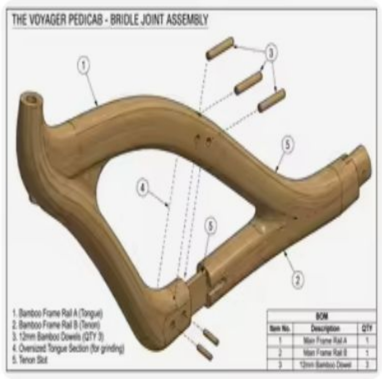
Tongue and Tenon Specification

- **Initial Build**: Extend the "tongue" side of the joint by an additional 50mm beyond the final meeting point.
- **Dowel Reinforcement**: Drill 12mm holes through the center of the joint once assembled. Use heat-treated bamboo dowels and waterproof structural epoxy (e.g., West System).
- **Grinding Phase**: After the epoxy has fully cured (24h), use a flap disc (60-80 grit) to grind the transition between pieces into a seamless, "liquid" curve as shown in the assembly diagrams.

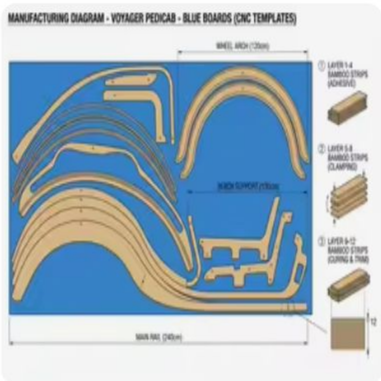
4. Technical Diagrams

Overall Dimensions & Views

Assembly & Joint Details



Template Cutting Layout



5. Compliance & Safety Analysis

- **Applicable Regulations**: Must comply with local municipal pedicab safety standards (e.g., Austin/New York Pedicab Ordinances).
- **Certification**: Requires static load testing (250kg) and dynamic brake testing.
- **Material Compliance**: Epoxy must be VOC-free for hospitality sector use.
- **Labeling**: Permanent chassis number must be engraved into the bamboo main rail BB-01.

Design Credit: This design was developed by **Accio.ai** for the Zenith Aero series.